SW Engineering CSC 648/848 Fall 2025

Gator Learn

Team #5

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Milestone 2 Part 1

Date submitted	Date revised
11/1/2025	-

1 Executive Summary

Ever had an experience where you thought "Oh no, I have an exam in a week, and I don't understand the material!" but don't know where to find support? Gator Learn has your back!! Gator Learn is a student-led academic support platform that's designed to empower San Francisco State University students through accessible, affordable and personalized tutoring exclusively for SFSU students, by SFSU students. Our platform fosters a collaborative learning environment built on shared academic experiences and community trust. We offer one-on-one and group tutoring across a wide range of SFSU courses including STEM, Humanities, Business, Writing, and general education requirements! Our mission is to enhance student success, confidence and retention by making academic support convenient and relatable. Sessions are available in-person and online with flexible scheduling to fit the demands of university life. Beyond coursework help, tutors provide study strategies, exam preparation, and mentorship to help students thrive long term!

Gator Learn! Helping Gators support Gators!

2 Data Items and Entities

Key Data Entities

1. **UserAccount**:

   Description: Stores basic information for every individual using the platform , including students, tutors, and admins.

   Attributes:

   1. Name
   2. E-mail
   3. Password
   4. Photo (Optional)

5. Description (Optional)
6. Pronouns (Optional)

Usage: Used for login, access control, and permission management. Acts as the foundational link between all user-related data (profiles, sessions, messages).

2. **TutorListing:**

Description: Represents a tutor's service offering

Attributes:

1. Student Name
2. Subject
3. Course(s)
4. Price per hour
5. Available time(s)
6. Resume (CV)
7. Description
8. Photo (Optional)
9. Tutoring Video Sample (Optional)
10. Live (True/False)

Usage: Displayed to students during tutor search and booking. It helps students to compare available tutors based on price, subject, and timing before making a reservation.

3. **Message:**

Description: A one-way communication channel where a student sends a message to a listing owner

Attribute:

1. Student Name
2. Listing Owner Name
3. Tutor Listing
4. Message Content
5. Date and Time
6. Phone Number (Optional)

Usage: Used by the listed student who is getting the tutoring request to get the information about the student, including phone number, and a message.

4. **Subject:**

   Description: Represents an SFSU academic subject that tutors can teach and students can search for.

   Attribute:

   1. Subject Name
   2. Category
   3. Description

   Usage: Used to categorize tutor listings and organize search filters by subject area. Supports subject-based matching during tutor listing browsing.

5. **Course:**

   Description: Represents an SFSU academic course that tutors can teach and students can search for

   Attribute:

   1. Course Name
   2. Subject
   3. Course Number
   4. Description

   Usage: Used to categorize tutor listings and organize search filters by course. Supports course-based matching during tutor listing browsing.

6. Feedback:

Description: Captures feedback from students after a tutoring session, including star rating and written comments.

Attribute:

1. Student Name
2. Listing Owner Name
3. Rating
4. Comment
5. Posted Date

Usage: Displayed on tutor profiles to help future students make informed decisions.

3 Functional Requirements

Priority 1

Unregistered Users

1. Unregistered users shall be able to register on a dedicated registration page with required name and @sfsu.edu email and optional photo, description, and pronouns
2. Unregistered users shall be able to look at home page, team page, and browsing page
3. Unregistered users shall be able to search and sort tutor listings by name, time, price, SFSU subject, and SFSU course number

Registered Users

4. Registered users shall inherit all unregistered user's requirements and permissions
5. Registered users shall be able to login on a dedicated login page
6. Registered users shall be able to logout
7. Registered users shall be able to one-way reply to tutor listings with an in-site messaging system.
8. Registered users shall be able to submit tutor listing applications with required title, description, available weekly times, price, and CV and optional photo and video for post approval
9. Registered users shall be able to use a dashboard that contains messages received or sent and pending or approved tutor listings.

Administration

10. Administrators shall inherit all registered user's requirements and permissions
11. Administrator accounts shall be assigned manually by website manager
12. Administrators shall be required to review all tutor listings before they go live

Priority 2

Registered Users

13. Registered users shall be able to edit account information after registering
14. Registered users shall be able to edit their tutor listings after approval.
15. Registered users shall be able to delete their tutor listings or applications.
16. Registered users shall be able to delete their messages

Administration

17. Administrators shall be able to remove any registered user or tutor listing

Priority 3

Unregistered Users

18. Unregistered users shall be able to view user pages
19. Unregistered users shall be able to get help from an FAQ/Help page

Registered Users

20. Registered users shall be able to reset their password
21. Registered users shall be able to report listings and messages
22. Registered users shall be able to provide feedback on listings

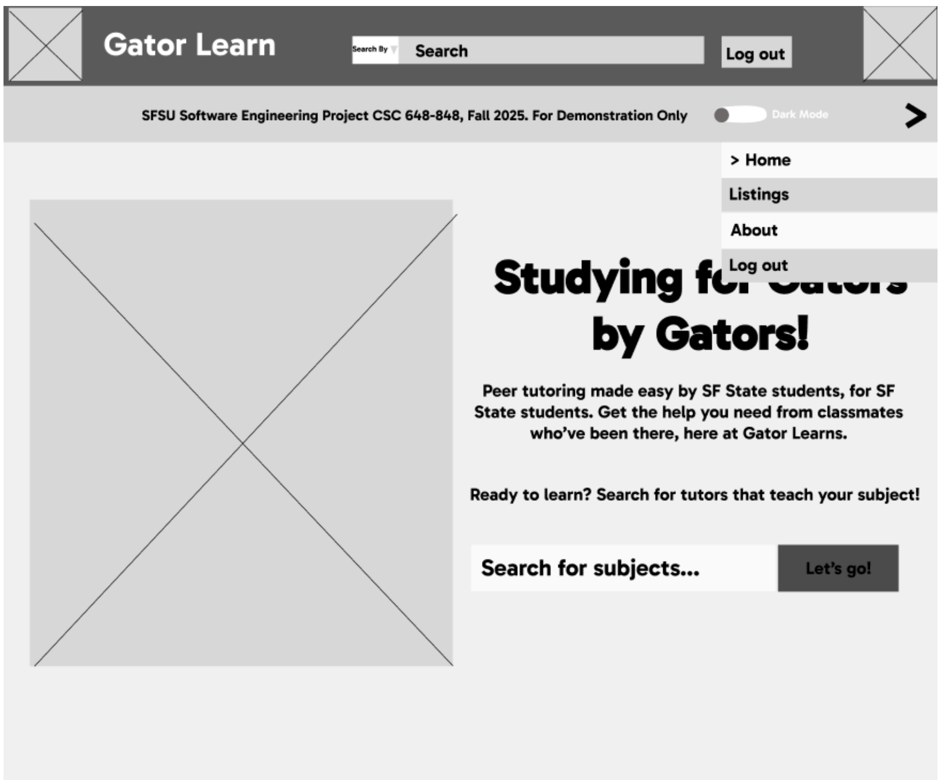
Administration

23. Administrators shall be able to view website analytics

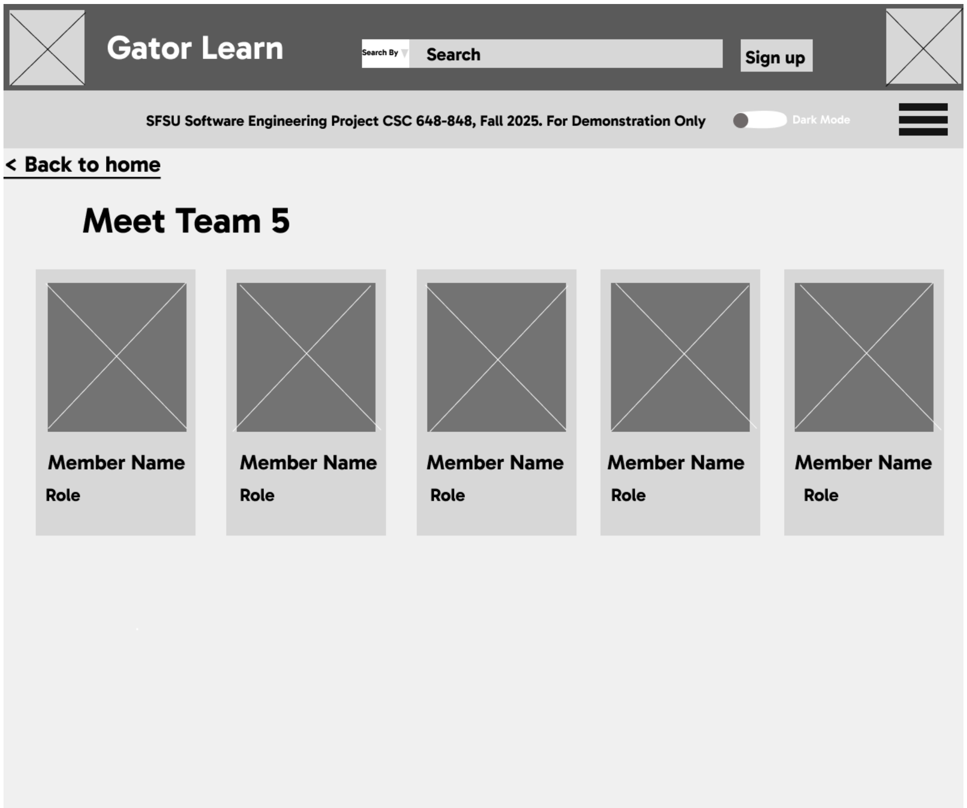
4 UI Storyboards

Use Case 1: Student browses and registers for site

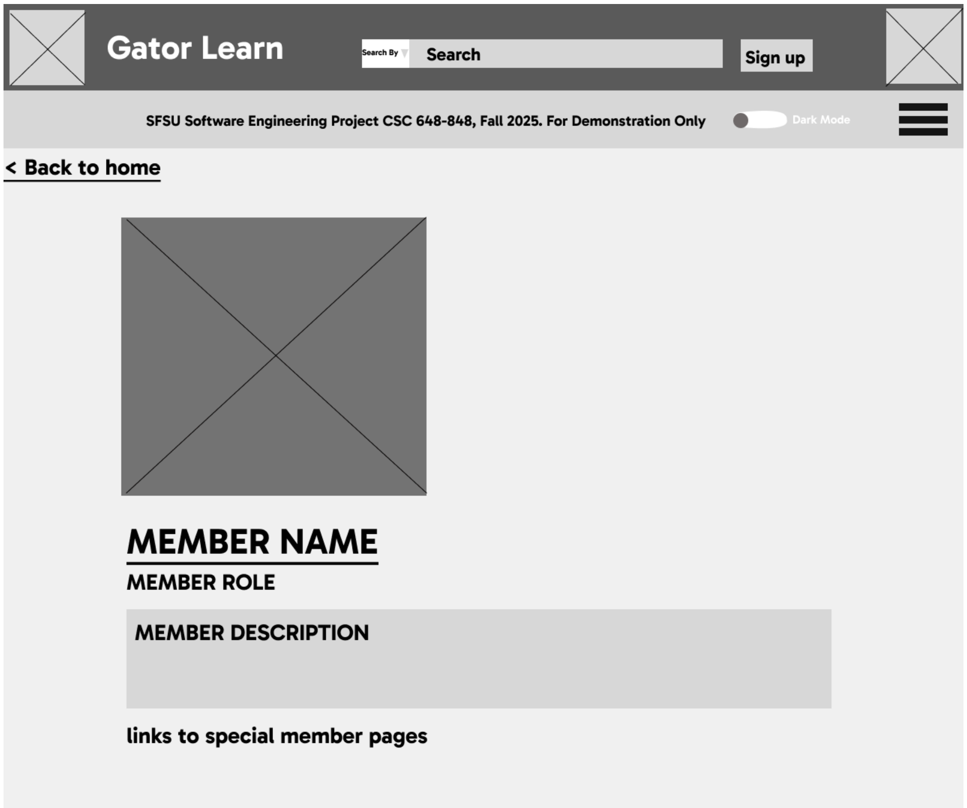
Alex scrolls through the site and browses all the pages available without registering.



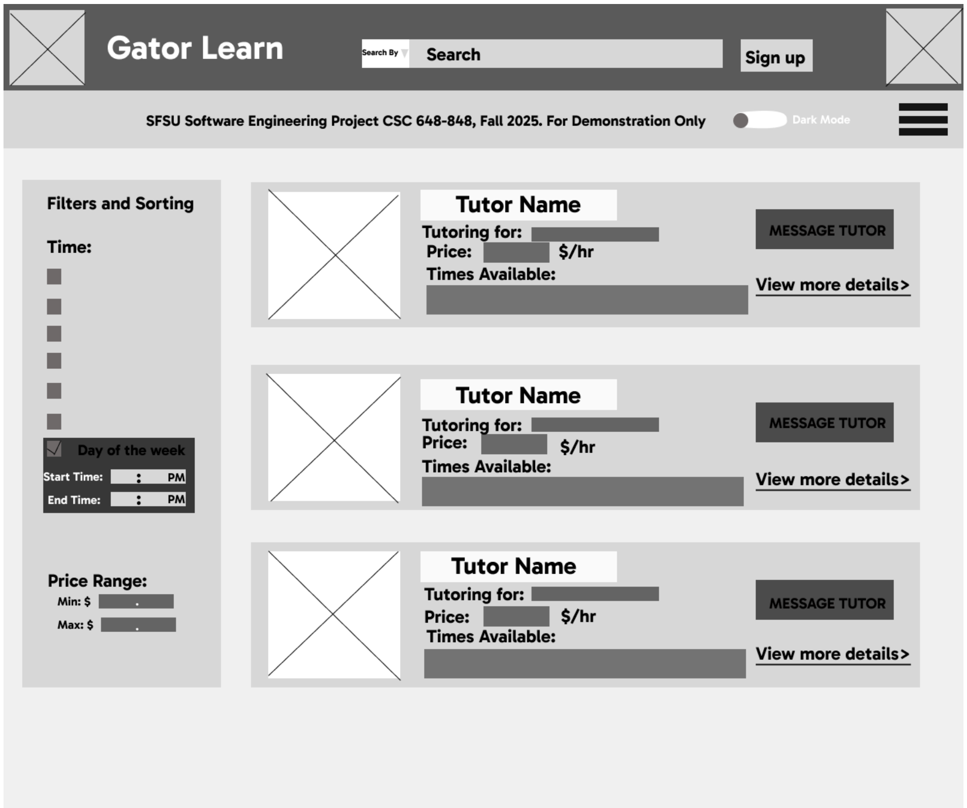
Alex sees that the site was made by SFSU students, too, and is able to click on the team page information without registering.



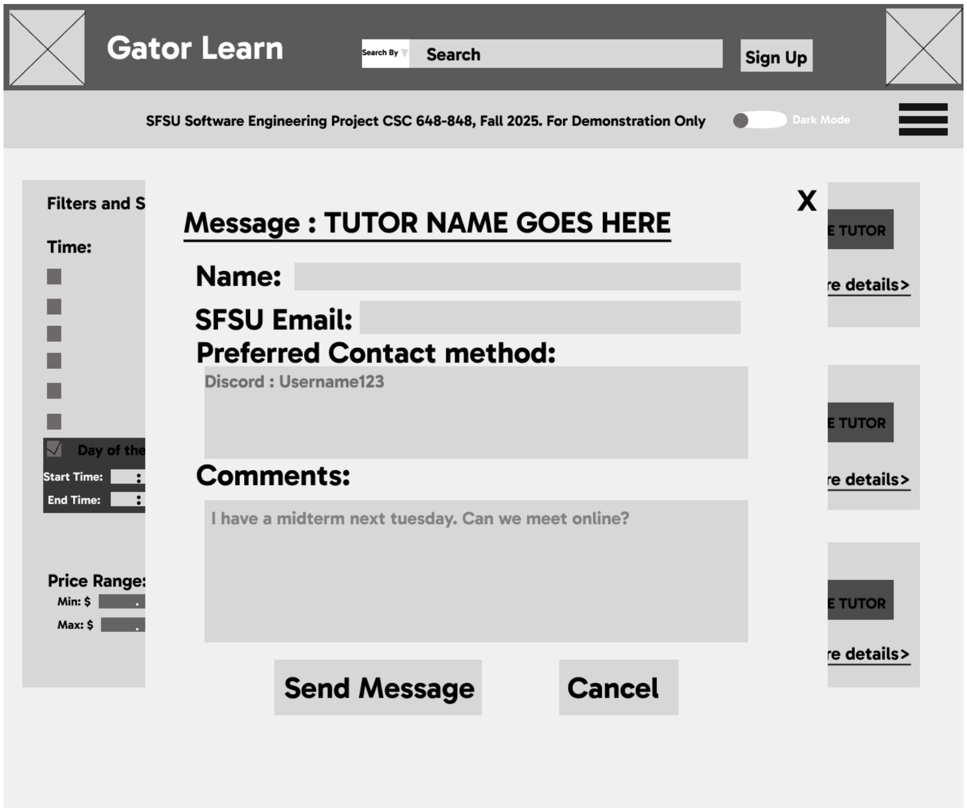
He clicks on one member's image and learns a bit more about them.



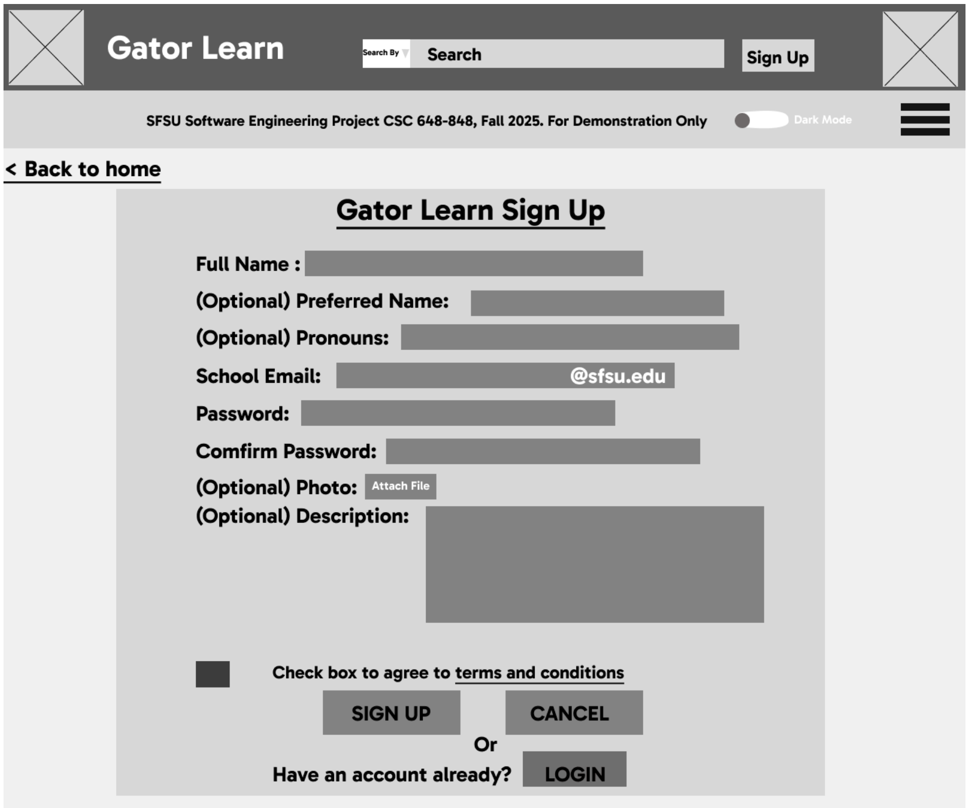
He changes pages through the drop down again and browses through the tutor listings.



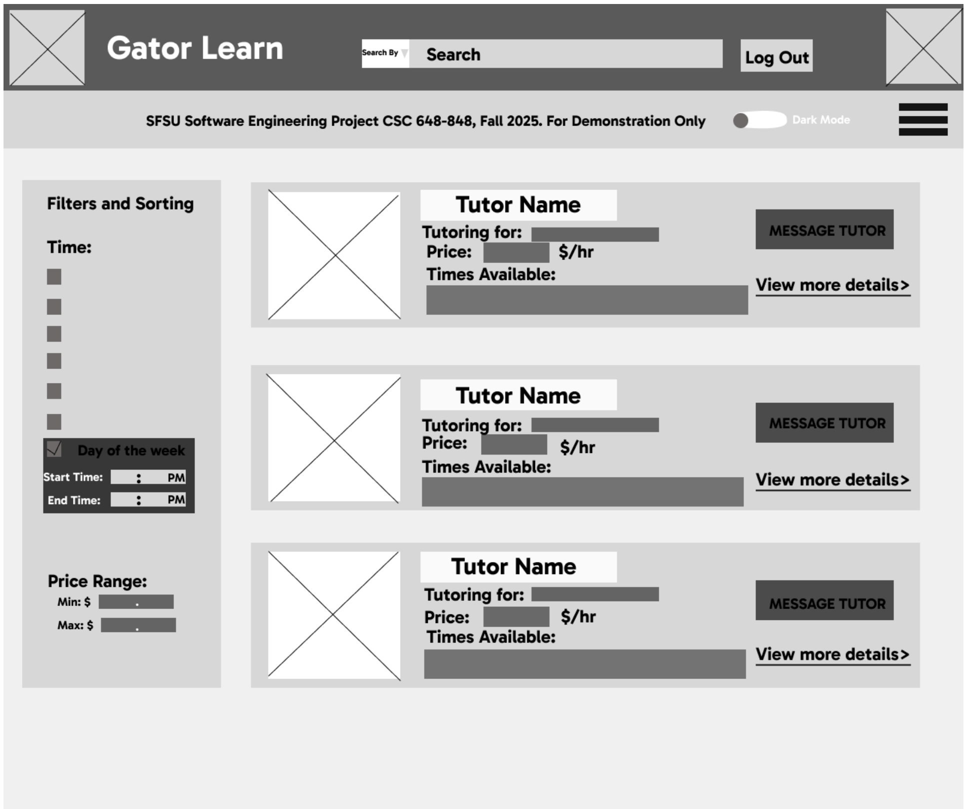
Alex tries to send a message without registering. The site sends him to sign up after he has written his message to the tutor.



Alex then signs up, and credentials are confirmed via SFSU email. His message gets sent with his sign up info.

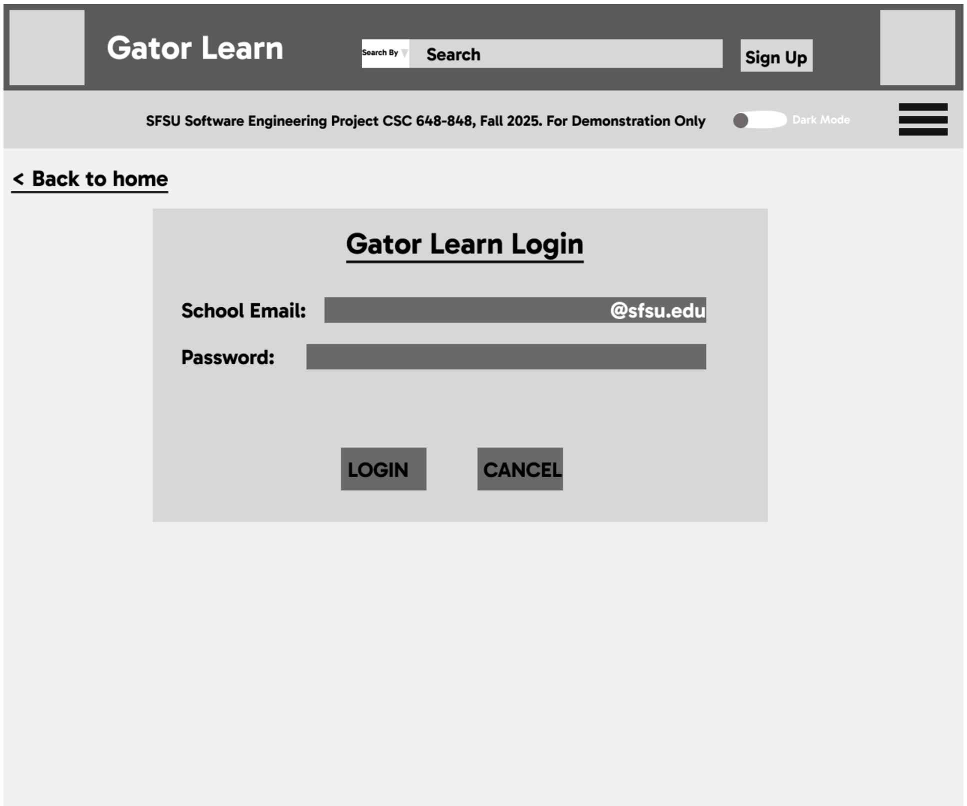


The site prompts Alex to his dashboard to see his message.

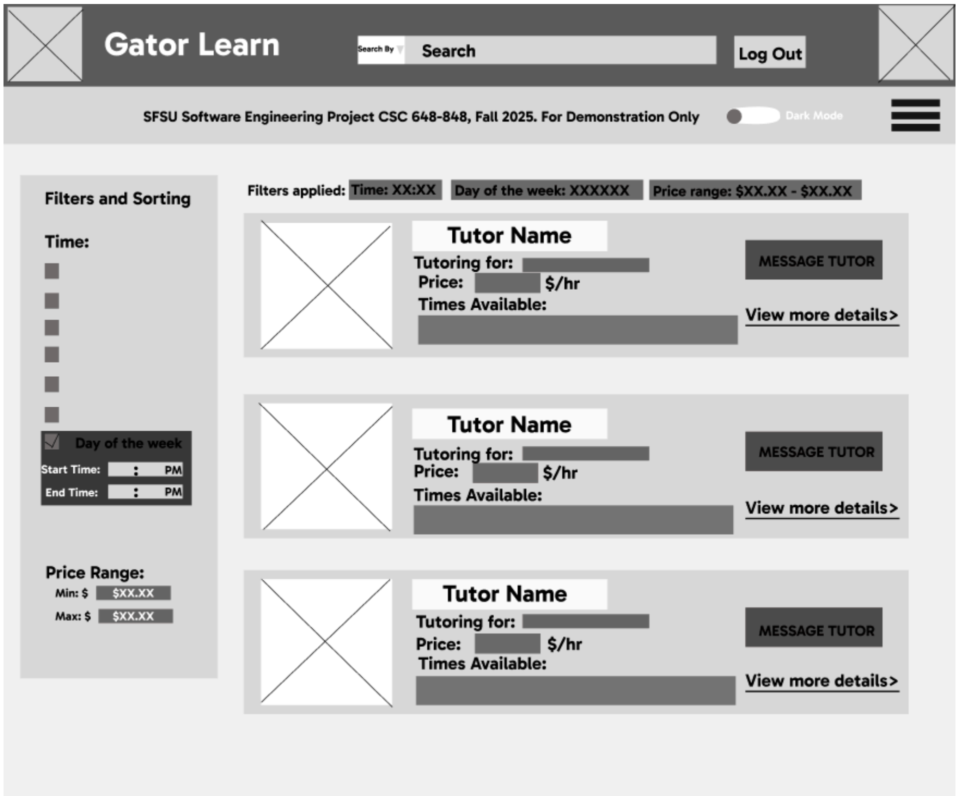

Gator Learn
Search By
Search
Log Out
SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only
Dark Mode
Filters and Sorting
Time:
Day of the week
Start Time: : PM
End Time: : PM
Price Range:
Min: $
Max: $
Tutor Name
Tutoring for:
Price:
$/hr
Times Available:
MESSAGE TUTOR
View more details>
Tutor Name
Tutoring for:
Price:
$/hr
Times Available:
MESSAGE TUTOR
View more details>
Tutor Name
Tutoring for:
Price:
$/hr
Times Available:
MESSAGE TUTOR
View more details>

Use Case 2: Student logs in and wants to message a tutor

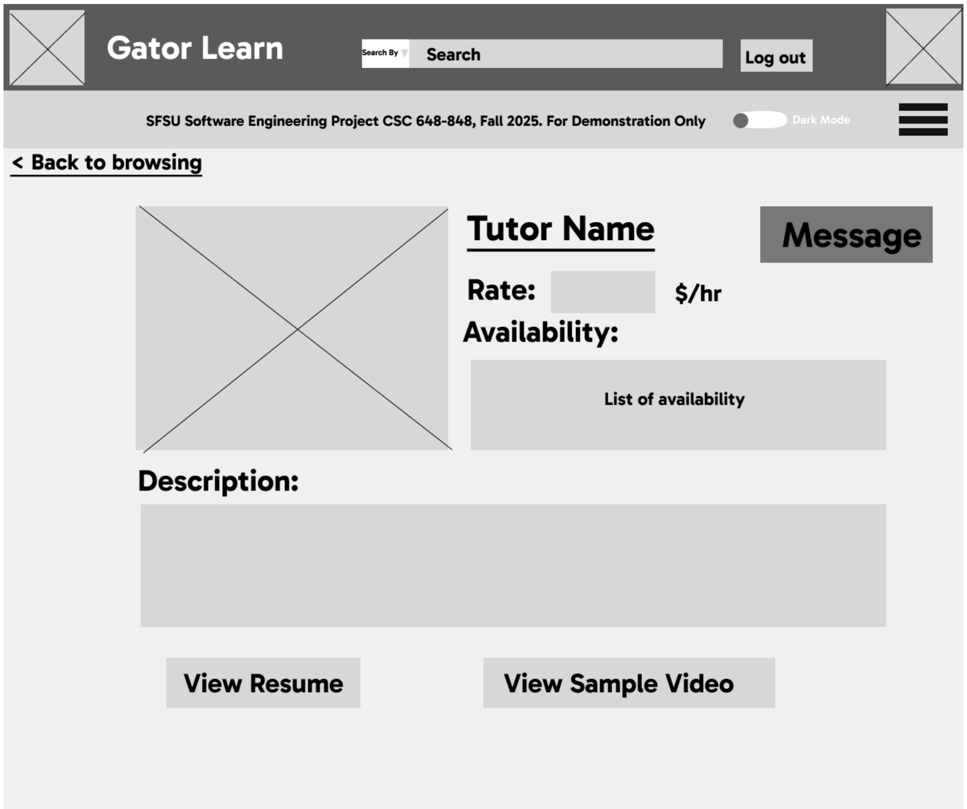
Sarah, an already registered user, logs back to the site to find another tutor for a different class



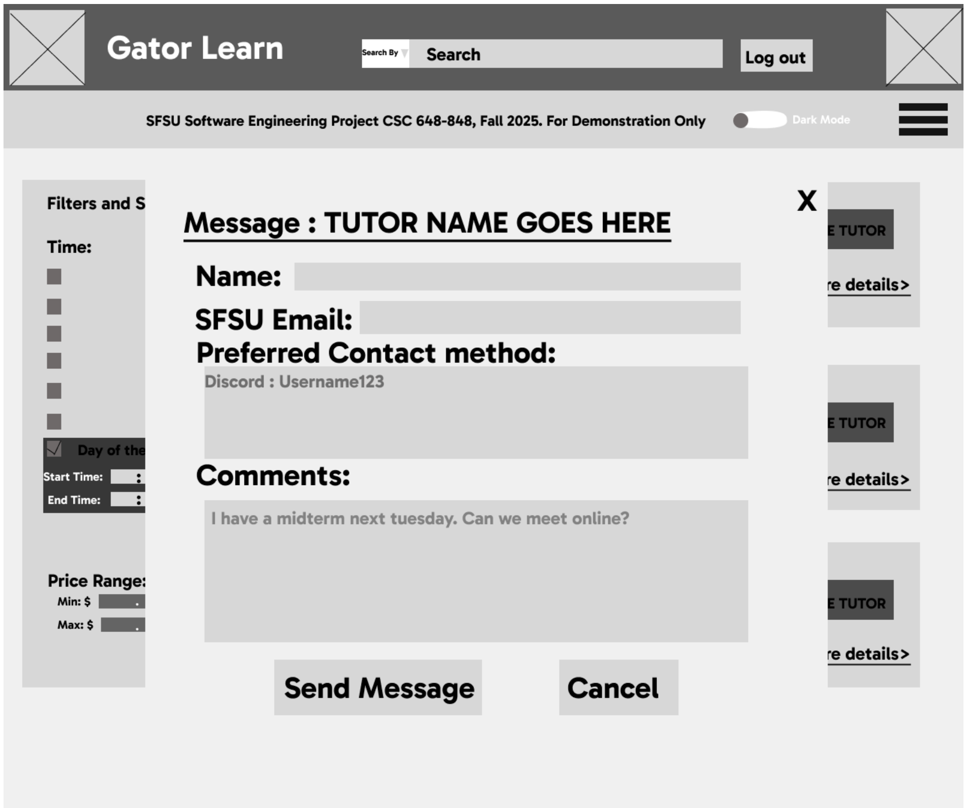
Sarah browses tutor listings, doesn't see something specific towards her, so she searches by her class and sorts by time availability



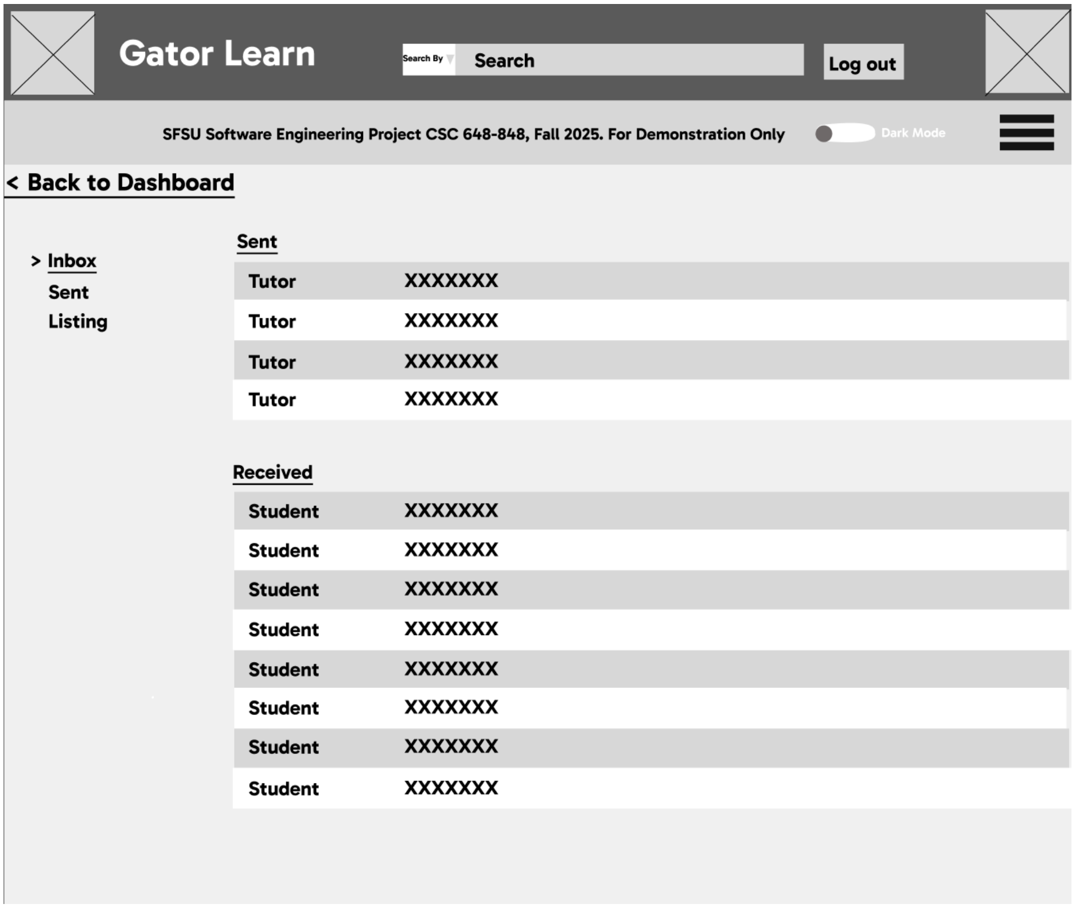
Sarah looks at the tutor listing page to see a bit more detail on the tutor.



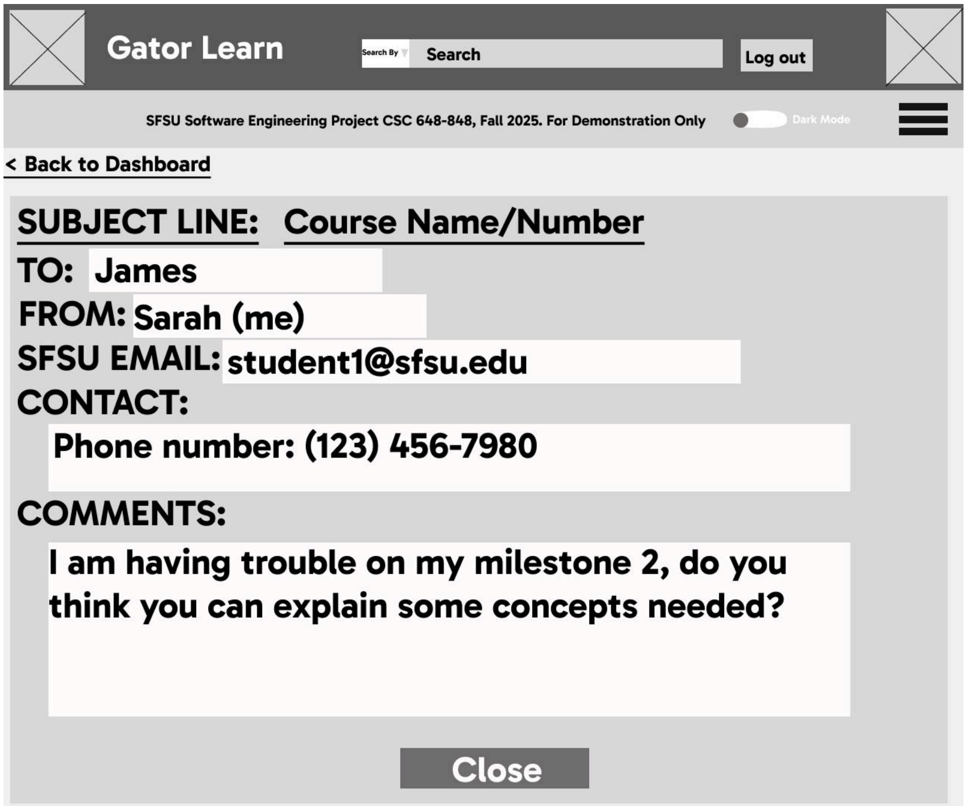
She thinks this tutor can help her so she presses the message tutor button. A form pops up to send a message to the tutor. She fills it out and sends it.



Sarah can then check her sent messages on her dashboard. Tutor contacts her off-site.

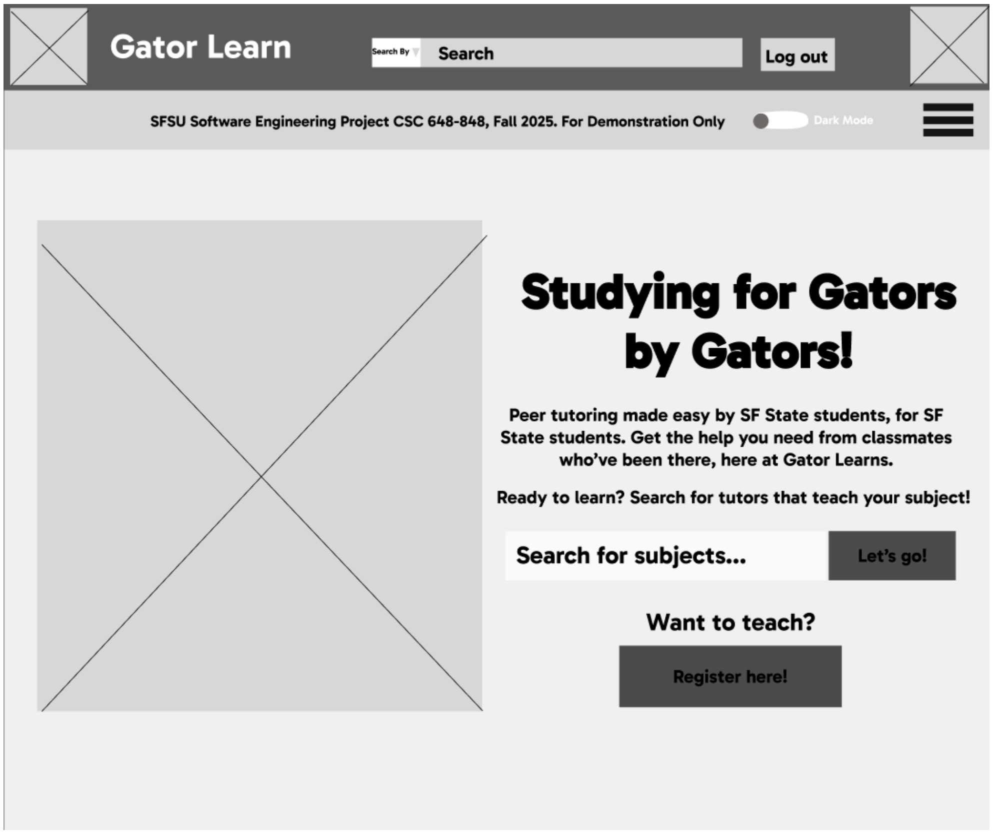


Sarah opens her message to the tutor.

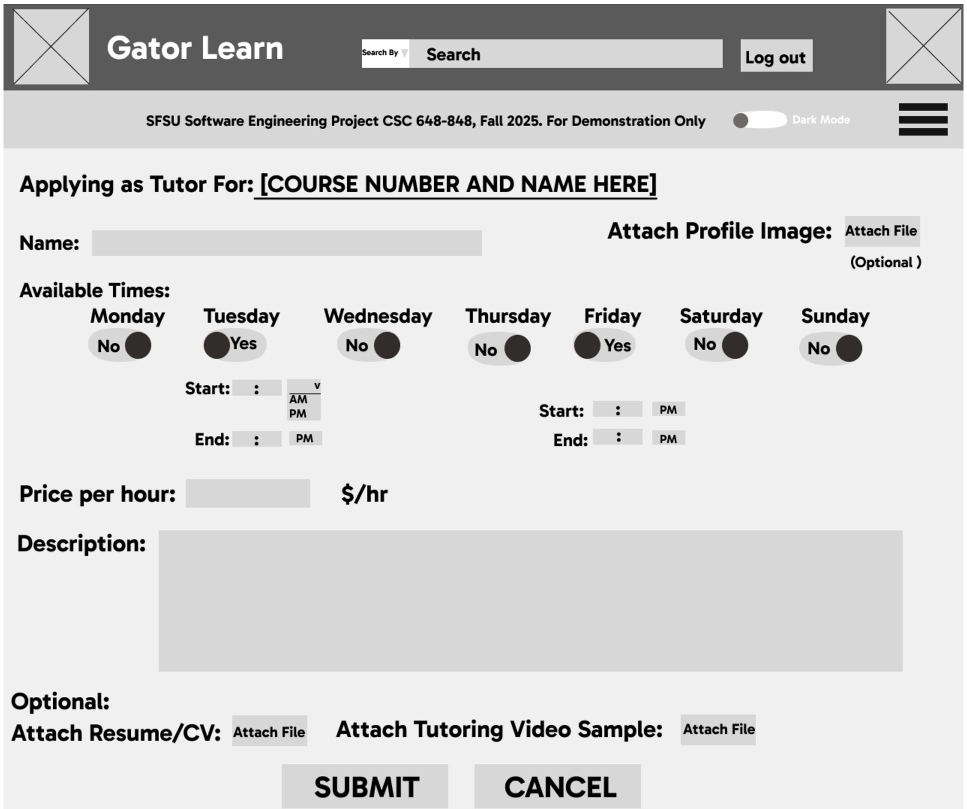

Gator Learn
Search By
Search
Log out
SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only
Dark Mode
< Back to Dashboard
SUBJECT LINE:
Course Name/Number
TO:
James
FROM:
Sarah (me)
SFSU EMAIL:
student1@sfsu.edu
CONTACT:
Phone number: (123) 456-7980
COMMENTS:
I am having trouble on my milestone 2, do you think you can explain some concepts needed?
Close

Use Case 3: Student Fills Out Tutor Listing Form

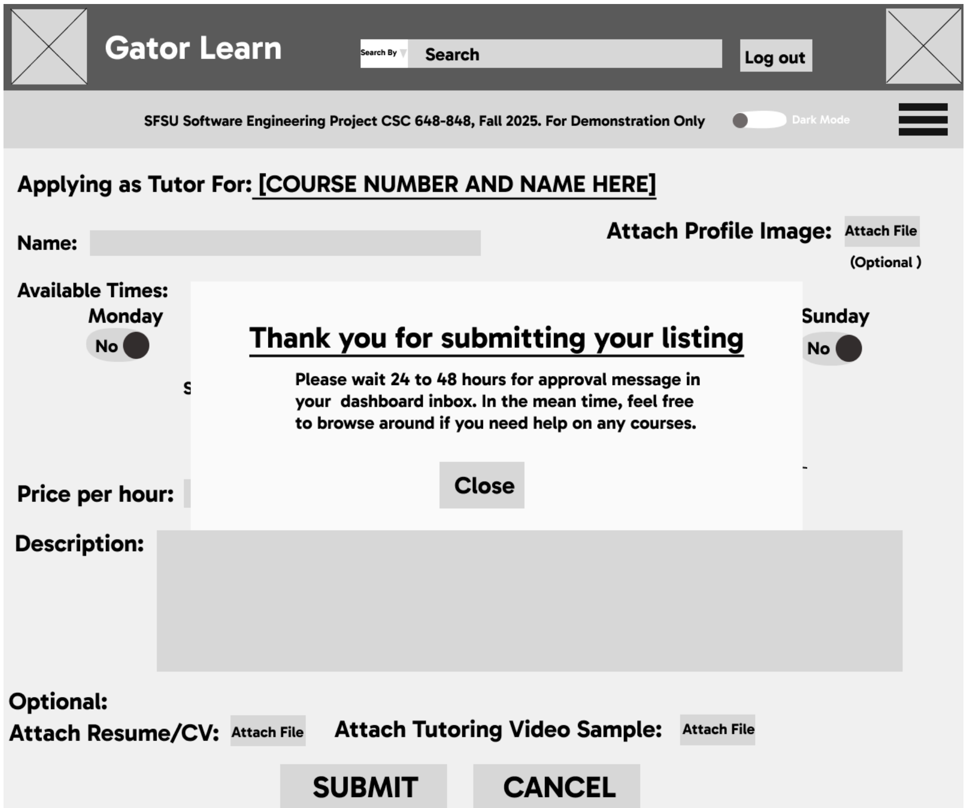
James signs up and browses around.



He sees that there are few tutors for a course he can teach, so he fills out a tutor listing form.



After James fills out his tutor listing, the site prompts him to wait a day or two for approval. He waits for approval.

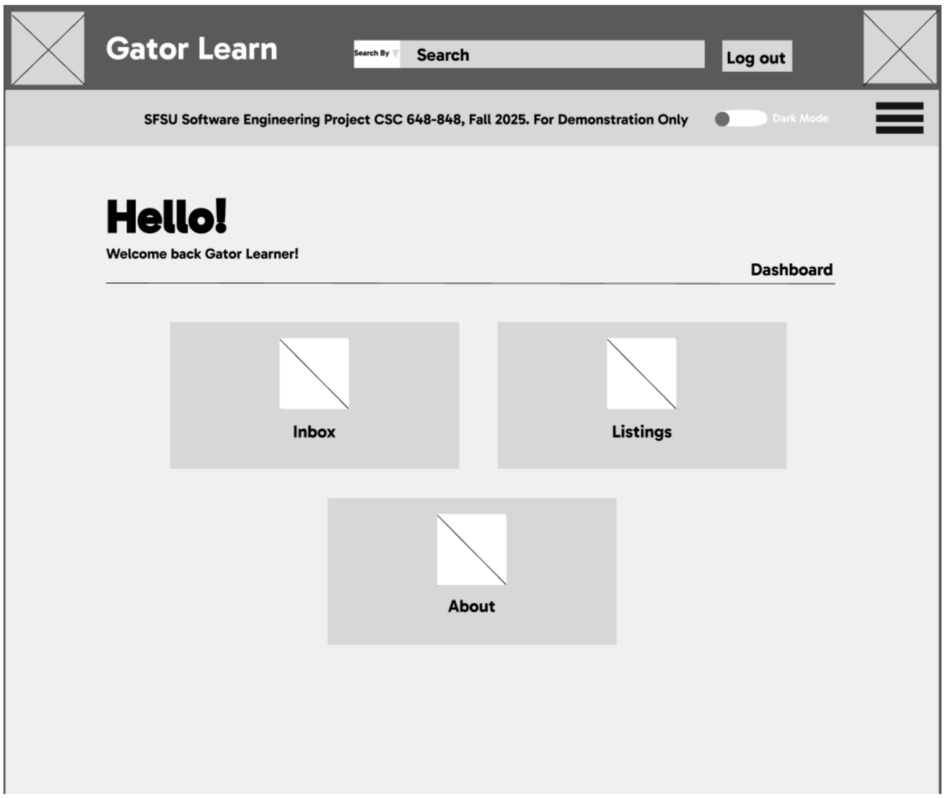


He logs out in the meantime.

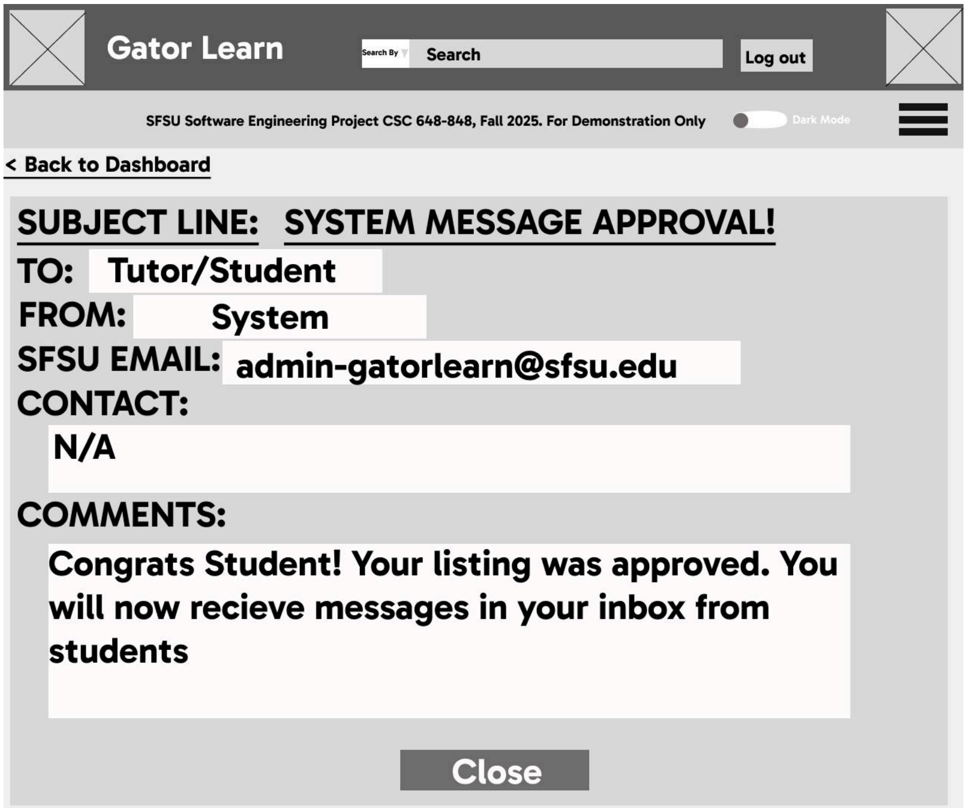

Gator Learn
Search By
Search
Log Out
SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only
Dark Mode
Filters and Sorting
Time:
Day of the week
Start Time: : PM
End Time: : PM
Price Range:
Min: $
Max: $
Tutor Name
Tutoring for:
Price:
Times Available:
MESSAGE TUTOR
View more details>
You are now logging out of Gator Learns.
Are you sure you want to log out?
(All unsaved work will be lost!)
Yes
No
Tutor Name
Tutoring for:
Price:
Times Available:
MESSAGE TUTOR
View more details>

Use Case 4: The Student can see the dashboard for replies on the tutor listing and send messages

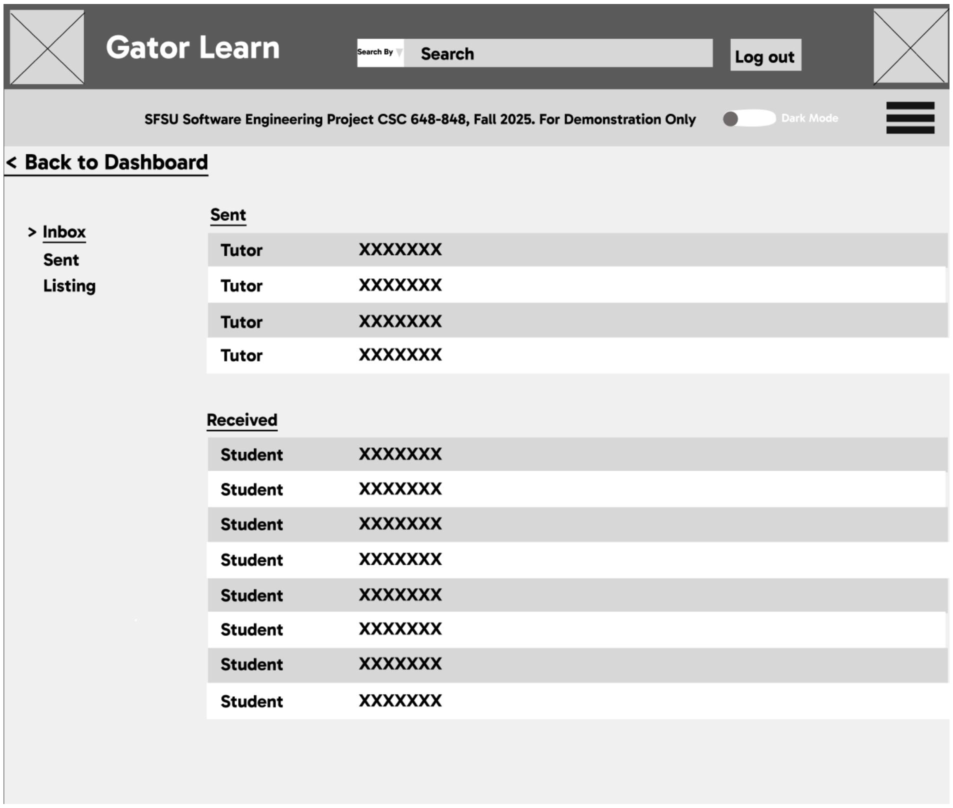
James logs back in and goes to his dashboard after waiting 2 days.



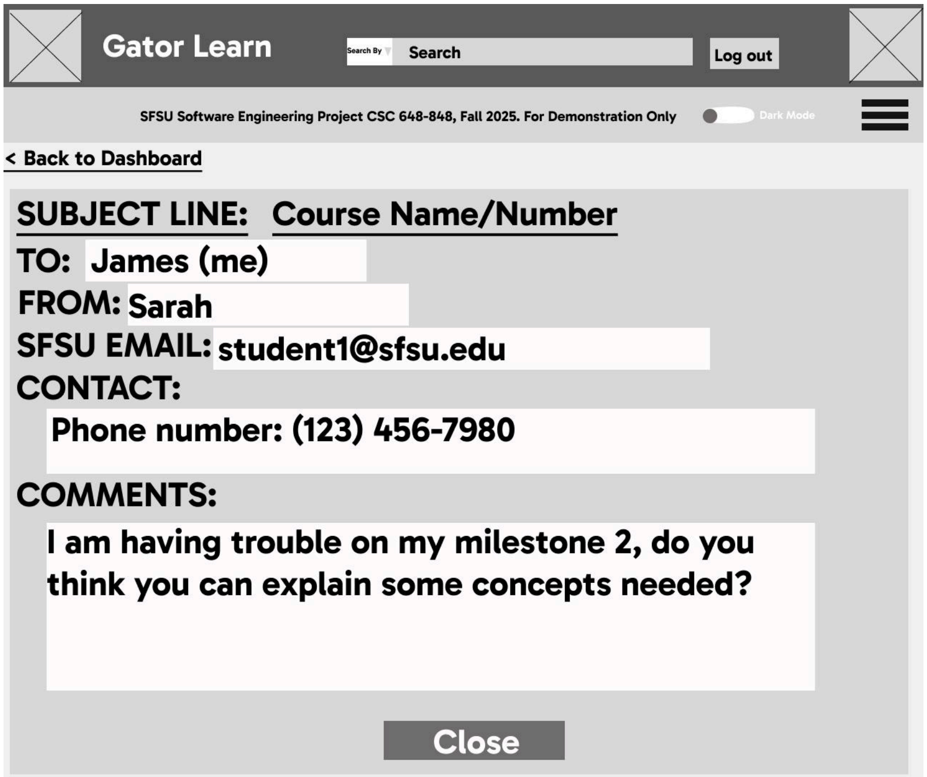
James sees he has a message from the system saying he was approved.



He also sees a few messages from students.



He clicks on one of the messages from the students and he chooses to read one and contact them off site.



Use Case 5: Admin will use Workbench for their responsibilities.

5 Architecture and Database Organization

Database Organization:

1. **UserAccount**
    a. userID (PK)
    b. name
    c. email
    d. password
    e. photo
    f. description
    g. pronouns

2. **TutorListing**
    a. listingID (PK)

   b. userID (FK)
   c. subject
   d. course
   e. price
   f. availableTime
   g. resume
   h. description
   i. tutoringVideoSample
   j. live

3. **Messages**
   a. messageID (PK)
   b. listingID (FK)
   c. userID
   d. message
   e. phoneNumber
   f. dateAndTime

4. **Subject**
   a. subjectID (PK)
   b. subjectName
   c. category
   d. description

5. **Courses**
   a. courseID (PK)
   b. subjectID (FK)
   c. courseName
   d. courseNumber
   e. description

6. **Feedback**
   a. feedbackID (PK)
   b. listingID (FK)
   c. userID (FK)
   d. rating
   e. comment
   f. date

Media Storage:

We chose to store all uploaded media (images, CVs, and videos) as files on the server and only keep their file paths or URLs in the database.
This approach reduces database size, improves performance, and makes it easier to serve or update media content through a file system

Searching or Filtering:

We shall conduct searches and filters of tutor listings using MySQL queries.
For text comparison, the simple MySQL `LIKE` operator shall often be enough, however for more complex queries, we shall use the MySQL `REGEXP` operator.
For numerical comparison we shall use relational operators.

Some examples:

```
SELECT listingID FROM tutorListings WHERE course LIKE 'CSC%'
```

Selects all listingIDs whose corresponding listing is for a course string containing ‘CSC’

```
SELECT * FROM tutorListings WHERE subject REGEXP '^.*Arts?.*$'
```

Selects all TutorListings containing ‘Art’ or ‘Arts’ in the subject

```
SELECT * FROM tutorListings WHERE price <= 20
```

Selects all TutorListings whose price is less than 20 dollars per hour

In addition, all database queries shall be done using parameterized queries to sanitize user input, avoiding possible SQL injection attacks.
Parameterized queries in Java can be implemented using [PreparedStatement](s), for example:

```
// DB recognizes the ? in this query as “Fill in a single value here”, so
additional arguments to the query cannot be appended
String queryStr = “SELECT * FROM tutorListings WHERE course LIKE ?”;

// Create the statement
PreparedStatement statement = sqlConnection.prepareStatement(queryStr);

// Set the first wildcard ‘?’ in the query to be userInputCourseStr
statement.setString(1, userInputCourseStr + “%”);

// Execute the query
ResultSet result = preparedStatement.executeQuery();
```

6 Key Risks

Skills Risks

Git Proficiency

Risk: Some team members are still getting comfortable with branching, merging, and avoiding conflicts, which can slow collaboration or cause lost work.

Resolution: Conduct a short Git workflow session, enforce pull-request reviews into develop and main, and share resources and tutorial videos for participants to watch.

Database Management

Risk: Limited experience with MySQL schema design and remote database deployment could cause performance or data integrity problems.

Resolution: Assign a dedicated database lead, research and watch tutorials and validate schema through local testing before deployment.

Schedule Risks

Meeting Commitments and Deadlines

Risk: Balancing the project workload with other coursework may lead to rushed deliverables near milestones.

Resolution: Use GitHub Projects for weekly task tracking, break milestones into smaller sections, and have check-ins to adjust workload distribution.

Technical Risks

Integrating All SFSU Courses and Majors

Risk: Populating an accurate list of all courses/majors may be time-consuming or technically challenging if scraped or manually entered.

Resolution: Start with a simplified subset (e.g., top departments), automate data retrieval with scripts if possible, and expand once basic functionality is stable.

Legal / Content Risks

Licensing and Third-Party Assets

Risk: Using UI libraries, icons, or course data without proper licensing could create copyright issues.

Resolution: Use only open-source MIT/BSD-licensed components and cite data sources (e.g., SFSU catalog).

7 Project Management

Our project is managed with the goal of meeting all deadlines, evenly distributing workload, and maintaining clear accountability across team members. We use **GitHub Projects** as our primary project management tool because it integrates directly with GitHub Issues and Pull Requests, making it simple to assign tasks, set deadlines, and track progress in one place. Each task is assigned to a responsible team member with a specific due date, and we conduct mid-point check-ins to ensure adherence to timelines.

Team communication occurs through **Discord**, where we have dedicated channels for updates, deadlines, and shared resources. We hold weekly one-and-a-half-hour calls (in addition to post-class meetings) to review progress, redistribute tasks as needed, and confirm our direction. Individual follow-ups are done when specific support or clarification is required.

For **Milestone 2**, we are using **Figma** to create wireframes and storyboards, enabling rapid prototyping and smooth transition to implementation. **Google Docs** is used for milestone documentation, providing real-time collaboration and version control that better suits group editing than binary Word files in Git. Following our instructor's guidance, the **frontend and**

backend teams work in parallel while coordinating on shared API interfaces to improve efficiency and reduce dependencies.

8 GenAI Usage

Identifying key issues in project (Milo, ChatGPT): **MEDIUM**

Wrote out several key issues on my own and then had ChatGPT elaborate and find others. I found that it lacked proper solutions for our project and missed context that was necessary for the solutions.

Formatting milestone and documentation (Milo, ChatGPT): **HIGH**

After completing sections I asked ChatGPT to reformat it without changing the text. Was usually very helpful and I liked the formatting and stylistic choices that it made.

9 Team Lead Checklist

1. *So far all team members are fully engaged and attending team sessions when required.*

   ISSUE: There have been a couple issues with completing deliverables on time or meetings missed, but we were able to resolve them internally.

2. *Team ready and able to use the chosen back and front end frameworks and those who need to learn are working on learning and practicing.*

   ON TRACK

3. *Team reviewed suggested resources before drafting Milestone 2.*

   ISSUE: We accidentally used Figma Sites instead of Figma Design, so our wire diagrams are low fidelity instead of high fidelity and we won’t be able to convert our storyboards into HTML. We plan on correcting this mistake in part 2 of Milestone 2.

4. *Team lead checked Milestone 2 document for quality, completeness, formatting and compliance with instructions before the submission.*

   DONE

5. *Team lead ensured that all team members read the final Milestone 2 document, and agree/understand it before submission.*

   ON TRACK

6. *Team shared and discussed experience with genAI tools among themselves.*

   DONE